KEVIN SLOTE

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EDUCATION

Georgia State University *2020 – July 2025*
Ph.D. in Applied Mathematics Atlanta, Georgia
Dissertation: *Causal Inference, Topological Data Analysis, and Machine Learning for Dynamics on Social and Neurological Networks*
Advisor: Igor Belykh **Georgia State University** *2013 – 2016*
M.S. in Abstract Algebra Atlanta, Georgia
Advisor: Florian Enescu **Georgia State University** *2009 – 2011*
B.S. in Mathematics and Statistics, Minor in Physics Atlanta, Georgia

PUBLICATIONS, PREPRINTS, PATENTS

Higher-Order Kolmogorov-Arnold Networks for Dynamics. Kevin Slote, Erik Bollt, Jeremie Fish. *ICLR 2027 Submitted* (2026)

Causation Under Emergence: Optimal Causation Entropy for Chaotic Oscillators. Kevin Slote, Jeremie Fish, Erik Bollt. Chaos (Erik Bollt Memorial Issue; Invited Submission, in final prep) (2026)

Deep-Koopman-KANDy: Dictionary Discovery for Deep-Koopman Operators with Kolmogorov-Arnold Networks for Dynamics. Kevin Slote, Erik Bollt, Jeremie Fish. *NeurIPS 26 Under Review* (2026)

KANDy: Kolmogorov-Arnold Networks and Dynamical System Discovery. Kevin Slote, Jeremie Fish, Erik Bollt. *SIADS Under Review* (2026)

CausationEntropy: Pythonic Optimal Causation Entropy. Kevin Slote, Jeremie Fish, Erik Bollt. *JMLR MLOSS Under Review* (2026)

Interictal Scale-Free Neural Dynamics Identify Epileptogenic Cortex. I Belykh, M Shamshad, K Smith, K Slote, M Dhamala. *Communications Medicine Under Review* (2026)

Node Degree Volatility for Seizure Onset Zone Localization.
Kevin J. Slote, Kelley M. Smith, Marrium Shamshad, Aditi Trivedi, Charles M. Epstein, Mukesh Dhamala, Igor Belykh. *Brain Communications Under Review* (2025)

How Advocacy Groups on Twitter and Media Coverage Can Drive U.S. Firearm Acquisition: A Causal Study.
Kevin Slote, Kevin Daley, Rayan Succar, Roni Barak Ventura, Maurizio Porfiri, Igor Belykh. *PNAS Nexus, 4(6), pgaf195* (2025)

Online Performance Estimation with Unlabeled Data: A Bayesian Application of the Hui-Walter Paradigm.
Kevin Slote, Elaine Lee. *Advances in Information and Communication, LNNS 1285, Springer* (2025)

Systems and Methods for Attacks, Countermeasures, Archiving, Data Leak Prevention, and Other Novel Services for Active Messages.
Simon P. Tyler, Nathaniel S. Borenstein, Jackie A. Maylor, Carlos Da Silva, Kevin Slote, Herb L. Roitblat. *U.S. Patent 12,001,544* (2024)

Systems and Methods for Attacks, Countermeasures, Archiving, Data Leak Prevention, and Other Novel Services for Active Messages.

Simon P. Tyler, Nathaniel S. Borenstein, Jackie A. Maylor, Carlos Da Silva, Kevin Slote, Herb L. Roitblat. *U.S. Patent Application 18/415,791* (2024)

OTHER PUBLICATIONS

Data center fatigue: Repurpose AI to solve the problems it creates. Kevin Slote. *Atlanta Journal-Constitution.* *(August 28, 2026)*

Evolving Network Dynamics Reveal Critical Nodes in the Epileptic Brain. Kelley Smith, Kevin Slote, Marrium Shamshad, Aditi Trivedi, Charles Epstein, Mukesh Dhamala, Igor Belykh. *SIAM News.* *(2026)*

Optimal Causation Entropy for the Next Generation. *SIAM News Blog, Society for Industrial and Applied Mathematics* *(2026)*

Causal Effects of Media and Twitter Advocacy on Firearm Purchases
Kevin Slote, Kevin Daley, Rayan Succar, Roni Barak Ventura, Maurizio Porfiri, and Igor Belykh.*SIAM News, Society for Industrial and Applied Mathematics* *(2025)*

AWARDS, MEDIA MENTIONS, & IMPACT

Clarkson Researchers Develop AI Tool That Finds the Equations Behind Complex Systems. Clarkson University, July 6, 2026. Coverage of KANDy and its application to interpretable equation discovery. Clarkson University

Researchers Develop AI Tool That Finds the Equations Behind Complex Systems. *Phys.org*, July 7, 2026. Phys.org

Clarkson Researchers Release Software to Discover Causal Patterns in Complex Data. Clarkson University, April 28, 2026. Feature on the CausationEntropy software project and its long-term maintenance. Clarkson University

14 Online Debates That Show How Social Media Divided the Nation, *INKl (2026)* Article Link

How Advocacy Groups on Twitter and Media Coverage Can Drive U.S. Firearm Acquisition: A Causal Study, Kevin Slote, Kevin Daley, Rayan Succar, Roni Barak Ventura, Maurizio Porfiri, and Igor Belykh. *PNAS Nexus*, Vol. 4, Issue 6, June 2025, pgaf195. Altmetric score: 41 (Top 5% for articles of its age). doi.org/10.1093/pnasnexus/pgaf195

14 Online Debates That Show How Social Media Divided the Nation. Travis Campbell. *The Free Financial Advisor (via inkl)*, November 3, 2025. inkl.com/news/14-online-debates-that-show-how-social-media-divided-the-nation

Outstanding Research Award Online Performance Estimation with Unlabeled Data: A Bayesian Application of the Hui-Walter Paradigm FICC (2025).

Georgia State Mathematicians Reveal Factors Driving Gun Sales in America, Amanda Head (J2025).

Mathematicians Reveal Factors Driving Gun Sales in America. Amanda Head (eds) Gaby Clark; reviewed by Andrew Zinin. *Phys.org*, July 10, 2025. phys.org/news/2025-07-mathematicians-reveal-factors-gun-sales.html

Gun Sales: Social Media & News Impact. Grace Chen. *Time.News*, Health Section, July 15, 2025. time.news/gun-sales-social-media-news-impact

How Social Media and News Drive Gun Sales. Amanda Head, Georgia State University. *Futurity*, July 15, 2025. futurity.org/gun-sales-news-social-media-3286772-2

Mathematicians Reveal Components Driving Gun Gross Sales in America. HealthMedicineNet, July 2025. healthmedicinet.com/hmn-2025-mathematicians-reveal-components-driving-gun-gross-sales-in-america/

RESEARCH INTERESTS

Non-linear Dynamical Systems, Topological Data Analysis, Data-driven Dynamical Systems, Algebraic Geometry, Ergodic Theory, Dynamical Systems, Machine Learning, Generative AI, Computational Topology, and Applied Category Theory.

RESEARCH EXPERIENCE

Clarkson Center for Complex Systems Science (C3S2) (August 2025 – Present)
Mentor: Erik Bollt, Clarkson University.
Non-linear dynamics, Kolmogorov-Arnold Networks. Developed open-source Optimal Causation Entropy (oCSE) software.

Biological and Engineering Networks Lab (Jan 2020 – July 2025)
Mentor: Igor Belykh, Georgia State University.
Non-linear dynamics and topological analysis of epileptology. Developed causal networks, topological data analytics, and machine learning methods for detecting the seizure onset zone in iEEG data, contributing to the prediction of surgical outcomes. Collaborated with neurologists to integrate findings into potential clinical workflows. Research details: Belykh Lab.

WeSafe Project (Jan 2020 – Present)
Mentor: Igor Belykh, Georgia State University
Analyzed causal links between Twitter advocacy, media narratives, and firearm acquisition trends using advanced causal network methods with Peter and Clark Momentary Conditional Independence (PCMCI) methods. The findings intend to inform discussions about regulatory policies and public safety initiatives. Research details: We-Safe.

Reductions of Noetherian Filtrations (Jan 2013 – Aug 2016)
Mentor: Florian Enescu, Georgia State University.
My research explored the reductions of Noetherian filtrations in commutative algebra and their applications to algebraic geometry and abstract algebra and touched on homological algebra and category theory. Developed theoretical results on Noetherian filtrations.

Principal Research Data Scientist (2017 – present)
Designed and implemented advanced machine learning models for natural language processing, including transformers. Applied causal inference and generative AI techniques to develop innovative solutions for predictive analytics and data security. Co-invented patents related to AI and security.

ACADEMIC SERVICE & LEADERSHIP

Science Café Speaker: Social Media and Firearms Clarkson University, Potsdam Civic Center Community Room, *(Fall 2025).*

Data-Driven Dynamics Seminar (Co-Organizer) Society for Industrial and Applied Mathematics (SIAM) Student Chapter, Georgia State University, *(2024).*

WORK EXPERIENCE

Research Associate (2025 – Present)
Clarkson Center for Complex Systems Science
Clarkson University, Potsdam, NY

Graduate Research Assistant (2020 – present)
Biological and Engineering Networks Lab
Georgia State University

Principal Research Data Scientist (2021 – 2025)
Generative AI, Transformers, Causal Models, Machine and Deep Learning
Egnyte, LLC, Mountain View, CA

Principal Research Data Scientist (2017 – 2021)
Generative AI, Transformers, Causal Models, Machine and Deep Learning
Mimecast, LLC, London, GB

PEER REVIEWER

IEEE Transactions on Network Science and Engineering *May 2021 – present*
PLoS Complex Systems *June 2025 – present*
Nature Scientific Reports *August 2026 – present*

HONORS AND SCHOLARSHIPS

Brains and Behavior Fellowship *2010*

Research Initiations in Mathematics, Mathematics Education and Statistics *2011*

INVITED TALKS, TUTORIALS, AND POSTERS

KANDy: Kolmogorov-Arnold Networks for Dynamics and Emerging Variants
Kevin Slote, Clarkson University.
CRUNCH Seminar, Division of Applied Mathematics, Brown University *(2026)*
Recorded Seminar

Attractor reconstruction and their topological data analysis for seizure onset detection
K Slote, I Belykh, M Dhamala, M Shamshad, K Smith.
Biology and Medicine Through Mathematics Conference *(2026)*
Recorded Seminar

Causal Network Discovery with the Open-Source CausationEntropy Software Library
Kevin Slote, Jeremie Fish, and Erik Bollt
Clarkson University; Texas State University *(2025)*

Social Media and Firearms
Kevin Slote, Clarkson University University *Science Cafe* *(2026)*

Node Degree Volatility for Seizure Onset Zone Identification
Kevin Slote, Georgia State University; Kelley Smith, Georgia State University; Marrium Shamshad, Georgia State University; Mukesh Dhamala, Georgia State University; Igor Belykh, Georgia State University. *SIAM Applied Dynamical Systems* *(2025)*

Aperiodic Spectral Neural Activity in Identifying Seizure Onset Zones
Marrium Shamshad, Georgia State University; Kelley Smith, Georgia State University; Kevin Slote, Georgia State University; Mukesh Dhamala, Georgia State University; Igor Belykh, Georgia State University. *SIAM Applied Dynamical Systems* *(2025)*

Hidden Geometry of Epilepsy: A Topological Biomarker for Seizure Onset Zone and Onset Localization
Kevin Slote, Georgia State University; Kelley Smith, Georgia State University; Marrium Shamshad, Georgia State University; Mukesh Dhamala, Georgia State University; Igor Belykh, Georgia State University. *Brains and Behavior* *(2025)*

The Twitter effect: How anti-regulation organizations drive firearm acquisitions in the United States
Kevin Slote, Georgia State University; Kevin Daley, US Naval Research Laboratory; Rayan Succar, New York University Tandon School of Engineering; Roni Ventura, New York University; Maurizio Porfiri, New York University Tandon School of Engineering; Igor Belykh, Georgia State University.
SIAM Applied Dynamical Systems *(2025)*

How advocacy groups on Twitter and media coverage can drive U.S. firearm acquisition: A causal study.
Contributed talk SIAM-SEAS, University of Tennessee (Mar 2025)

Dynamics of Brain Functional Networks and Epileptic Seizures
Slote Kevin, Georgia State University; Kelly Smith, Georgia State University; Marrium Shamshad, Georgia State University; Mukesh Dhamala and Igor Belykh, Georgia State University. *Brains and Behavior* *(2021)*

Reductions of Noetherian Filtrations
Kevin Slote, Georgia State University. *Abstract Algebra Seminar* *(2015)*

Random Walks on Weakly Hyperbolic Groups
Kevin Slote, Georgia State University. *Big Ideas in Dynamics* *(2023)*

The Twitter effect: How anti-regulation organizations drive firearm acquisitions in the United States
Kevin Slote, Georgia State University; Kevin Daley, US Naval Research Laboratory; Rayan Succar, New York University Tandon School of Engineering; Roni Ventura, New York University; Maurizio Porfiri, New York University Tandon School of Engineering; Igor Belykh, Georgia State University. *SIAM Mathematical Data Science* *(2024)*

The Twitter effect: How anti-regulation organizations drive firearm acquisitions in the United States
Kevin Slote, Georgia State University; Kevin Daley, US Naval Research Laboratory; Rayan Succar, New York University Tandon School of Engineering; Roni Ventura, New York University; Maurizio Porfiri, New York University Tandon School of Engineering; Igor Belykh, Georgia State University. *Georgia Scientific Computing Conference* *(2025)*

Data-Driven Modelling of How Disinformation and Conspiracy Theories Propagate in Social Networks
Kevin Daley and Kevin Slote, Georgia State University; Maurizio Porfiri, New York University Tandon School of Engineering; Igor Belykh, Georgia State University. *SIAM Conference on Dynamical Systems (2023)*

A Functional Network Criterion for Identifying Seizure Onset Zones from iEEG Recordings abstract
Smith Kelley, Georgia State University; Kevin Slote, Georgia State University; Marrium Shamshad, Georgia State University; Mukesh Dhamala and Igor Belykh, Georgia State University. *SIAM Mathematical Data Science* *(2024)*

Dynamics of Brain Functional Networks and Epileptic Seizures
Slote Kevin, Georgia State University; Kevin Daley, Georgia State University; Kelly Smith, Georgia State University; Mukesh Dhamala and Igor Belykh, Georgia State University. *Brains and Behavior (2021)*

Stopping the Spread of Disinformation in Social Networks with Real-Time Control: a Data-Driven Case Study of Online COVID-19 Anti-Vaccine Conspiracy Theories
Kevin Daley and Kevin Slote, Georgia State University; Maurizio Porfiri, New York University Tandon School of Engineering; Igor Belykh, Georgia State University. *SIAM Applied Dynamical Systems (2021)*

PROFESSIONAL MEMBERSHIPS

Mathematical Association of America (MAA) *2022 – Present*
Society for Industrial and Applied Mathematics (SIAM) *2020 – Present*
Special Interest Group on Knowledge Discovery and Data Mining (SigKDD) *2019*

TECHNICAL SKILLS

Programming Languages	Python, Java, C++, Javascript, Bash, R, Haskell
Tools	PyTorch, Tensorflow, Keras, CUDA, MATLAB, Stan
Software	LaTeX, Git, Mathematica, Lean